

WEEKLY JOURNAL CLUB · YOU RECORD · THE PAPER DOES THE TALKING



AI in Medicine Paper Club

Episode 1 — When the LLM reaches the clinic

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THIS WEEK'S PAPER

A generative LLM in live primary care

Agweyu, Mwaniki, Menon, Korom et al.

Nature Medicine **32**, 3032–3039 (published 26 June 2026)

GPT-4o, inside the EMR of 16 Kenyan clinics. ~10,000 patients. A hard clinical endpoint.

- Open access, **CC BY 4.0**
- Pan-African Clinical Trials Registry: 202502499779176
- Protocol: zenodo.org/records/15788148

WHY THIS PAPER

Exams are not clinic

- LLMs look strong on vignettes and licensing-style tests
- **60%** of amenable deaths in LMICs occur after people already reached care
- Kenya: much of primary care is delivered by **clinical officers** (3-year diploma)
- Sub-Saharan Africa: ~**0.3 physicians / 1,000** people

If a copilot helps here, that is bigger than another AUC. If it does not, we should know.

THE INTERVENTION

AI Consult 2.0 — GPT-4o in the EMR

Both arms

- Same cloud EMR
- Usual Kenyan guidelines
- No extra pay for adherence

Intervention only

- GPT-4o (May 2025), temperature 0.1
- Reads structured + free text
- Identifiers stripped
- Green / yellow / **red** overlay
- Accept, edit, or ignore

What this is not

- Not a chatbot the patient sees
- Not GPT versus doctors on a quiz
- Not enforced use

US\$0.04

MEAN MODEL COST PER PATIENT

DESIGN

Pragmatic cluster RCT

16

PENDA HEALTH CLINICS

103

CLINICAL OFFICERS

9,347

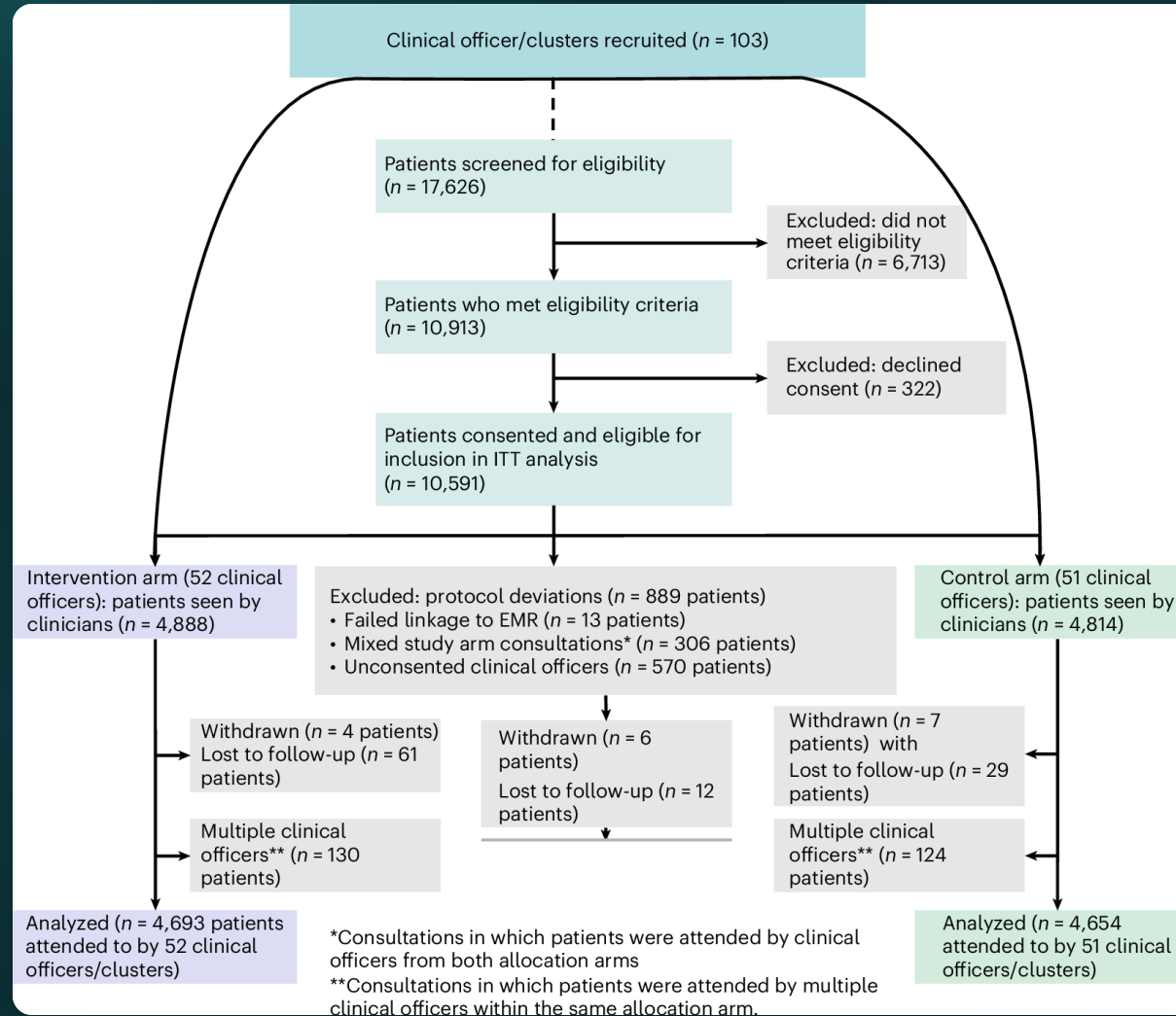
ENCOUNTERS ANALYSED

14 d

TREATMENT-FAILURE
WINDOW

- Randomized at **clinician** level (52 LLM / 51 control) — Nairobi & Kiambu, 22 Apr–16 Jul 2025
- Primary: expert-adjudicated **treatment failure** (return, escalation, missed dx, unsafe Rx, death)
- Kenyan family-physician panel, blinded to allocation
- Powered for a **50% relative drop** (2% → 1%)

Fig. 1 — CONSORT: clusters, then patients



Screened 17,626 → analysed 4,693 intervention vs 4,654 control. Most visits: adults 18–55, 56% women, ~60% febrile/infectious.

PRIMARY OUTCOME

Treatment failure did not fall

2.2%

LLM ARM · 102 / 4,693

2.0%

CONTROL · 94 / 4,654

0.77

AOR · 95% CI 0.55–1.08

0.13

P VALUE · ITT = PP

- Covariate-adjusted aOR **0.72** (0.50–1.03, *P* = 0.07) — still not significant
- Bayesian risk difference: **–5 failures / 1,000** (95% CrI –13 to +1)

They can reject a revolution. They cannot confirm a nudge. That is an honest null.

Table 2 — ITT and per-protocol sit on the same number

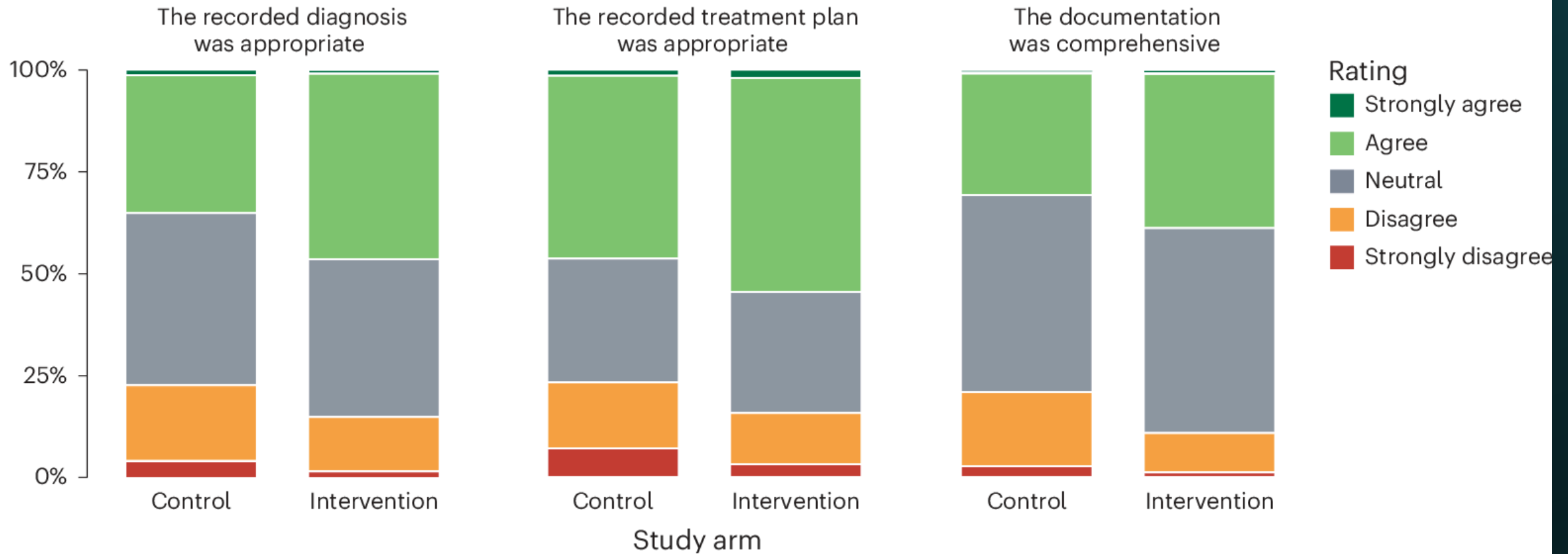
Table 2 | Treatment failure by study arm (intention-to-treat and per-protocol analyses)

Analysis	Characteristic	Failure, <i>n</i> (%)	No failure, <i>n</i> (%)	aOR	95% CI	Pvalue
ITT (adjusted for facility and clinical officer)	Control	94 (2.0)	4,560 (98.0)	0.77	0.55 to 1.08	0.13
	Intervention	102 (2.2)	4,591 (97.8)			
Per protocol (adjusted for facility and clinical officer)	Control	93 (2.0)	4,528 (98.0)	0.77	0.55 to 1.08	0.13
	Intervention	102 (2.2)	4,591 (97.8)			
ITT (cluster and covariate adjusted*)	Control	94 (2.0)	4,560 (98.0)	0.72	0.50 to 1.03	0.07
	Intervention	102 (2.2)	4,591 (97.8)			

Failure counts 94 vs 102. Odds ratio below 1; confidence interval crosses 1.
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Fig. 2a — Notes got better

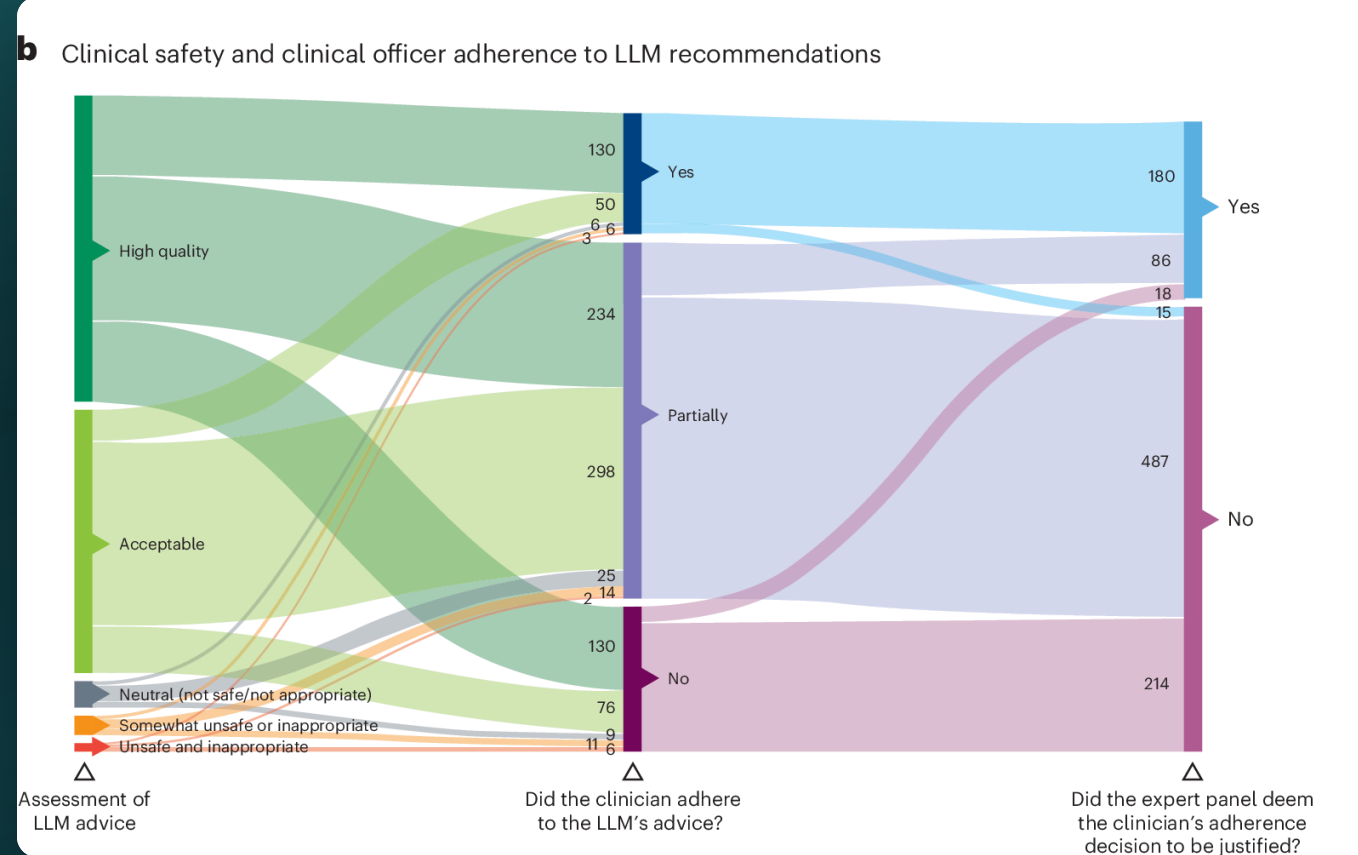
a Quality of the final clinical documentation among control and LLM-assisted consultations by domain



Appropriate diagnosis aOR 1.74 · comprehensive note 1.68 · appropriate plan 1.71 — all $P < 0.001$.

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Fig. 2b — Linger here: a mostly-safe model, a messy team



1,000 red alerts: 91.8% definitely/mostly safe · 1.1% unsafe. Followed fully 19.5% / partly 57.3% / not at all 23.2%. Expert panel: follow-or-ignore **not justified in 71.6%**.

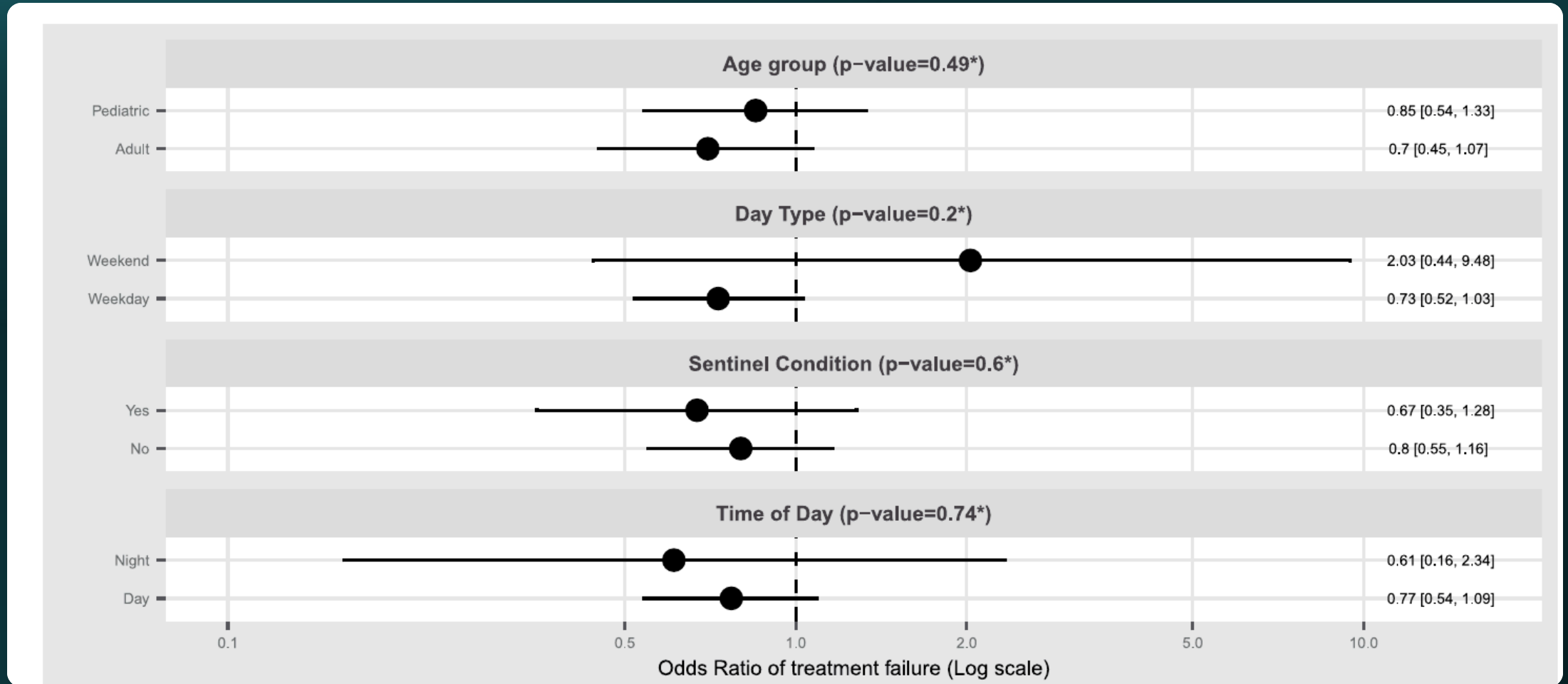
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Fig. 3 — Prescribing and sentinel conditions, mostly null

Prescription: antibiotics Received treatment ≥38.0 °C and prescribed antibiotics: ➤ 135 (of 170; 79%) in control arm ➤ 110 (of 145; 76%) in intervention arm Received proper treatment (including withholding): ➤ 79 (of 170; 46%) in control arm ➤ 61 (of 145; 42%) in intervention arm	Received poor-quality/incorrect treatment Patients for whom antibiotic use was appropriate, but poor-quality/unsafe prescriptions were identified: ➤ 14 (of 48; 29%) in control arm ➤ 5 (of 29; 17%) in intervention arm	Effect of the intervention Correct antibiotic use was similar between groups (OR 0.86; 95% CI 0.48 to 1.55; P = 0.60)
Prescription: antimalarials Received proper diagnosis Of patients who received an antimalarial prescription... ➤ 23 (of 29; 79%) in control arm ➤ 26 (of 30; 87%) in intervention arm ...had a malaria test	Received poor-quality/incorrect treatment Patients who received an antimalarial prescription despite a negative test or with no test performed: ➤ 7 (24%) in control arm ➤ 6 (20%) in intervention arm	Effect of the intervention The intervention did not reduce incorrect antimalarial use compared with control (OR 0.76, 95% CI 0.17 to 3.43, P = 0.70)
Diagnosis: acute malnutrition Received proper diagnosis Children under 5 years assessed and diagnosed with malnutrition: ➤ 43 (of 120; 36%) in control arm ➤ 49 (of 130; 38%) in intervention arm	Received appropriate referral Eligible children referred to a nutritionist: ➤ 35 (of 120; 29%) in control arm ➤ 44 (of 130; 34%) in intervention arm	Effect of the intervention The odds of receiving a malnutrition diagnosis did not differ between study arms (OR 0.91, 95% CI 0.50 to 1.64, P = 0.70) Referral to a nutritionist was also similar across study arms. (OR 1.14, 95% CI 0.61 to 2.13, P = 0.70)
Diagnosis: type 2 diabetes mellitus Symptoms identified Patients classified as at risk for diabetes: ➤ 1,578 (of 2,896; 54%) in control arm ➤ 1,447 (of 2,904; 50%) in intervention arm Received proper investigations Identified as at risk for diabetes and tested: ➤ 92 (of 1,578; 5.8%) in control arm ➤ 74 (of 1,447; 5.1%) in intervention arm	Received proper treatment Of those patients diagnosed with diabetes... ➤ 25 (of 33; 76%) in control arm ➤ 24 (of 32; 75%) in intervention arm ...started treatment	Effect of the intervention Those classified as at risk for diabetes were lower in the intervention arm than in the control arm (OR 0.88, 95% CI 0.78 to 0.98, P = 0.023) Among those identified as at risk, diabetes testing was infrequent and comparable between groups (OR 0.84, 95% CI 0.56 to 1.26, P = 0.40) Treatment initiation among those diagnosed with diabetes was also similar in both groups (OR 0.80, 95% CI 0.01 to 59.3, P > 0.90)
Diagnosis: hypertension Received diagnosis Patients who received a new hypertension diagnosis: ➤ 153 (of 2,783; 5.5%) in control arm ➤ 132 (of 2,799; 4.7%) in intervention arm	Received proper treatment Patients with a new hypertension diagnosis who started treatment: ➤ 36 (of 153; 23.5%) in the control arm ➤ 23 (of 132; 17.4%) in the intervention arm	Effect of the intervention Odds of receiving a new hypertension diagnosis were comparable across the control and intervention arms (OR 0.85, 95% CI 0.67 to 1.08, P = 0.20) Among patients newly diagnosed with hypertension, treatment initiation was also similar across arms (OR 0.68, 95% CI 0.37 to 1.23, P = 0.20)

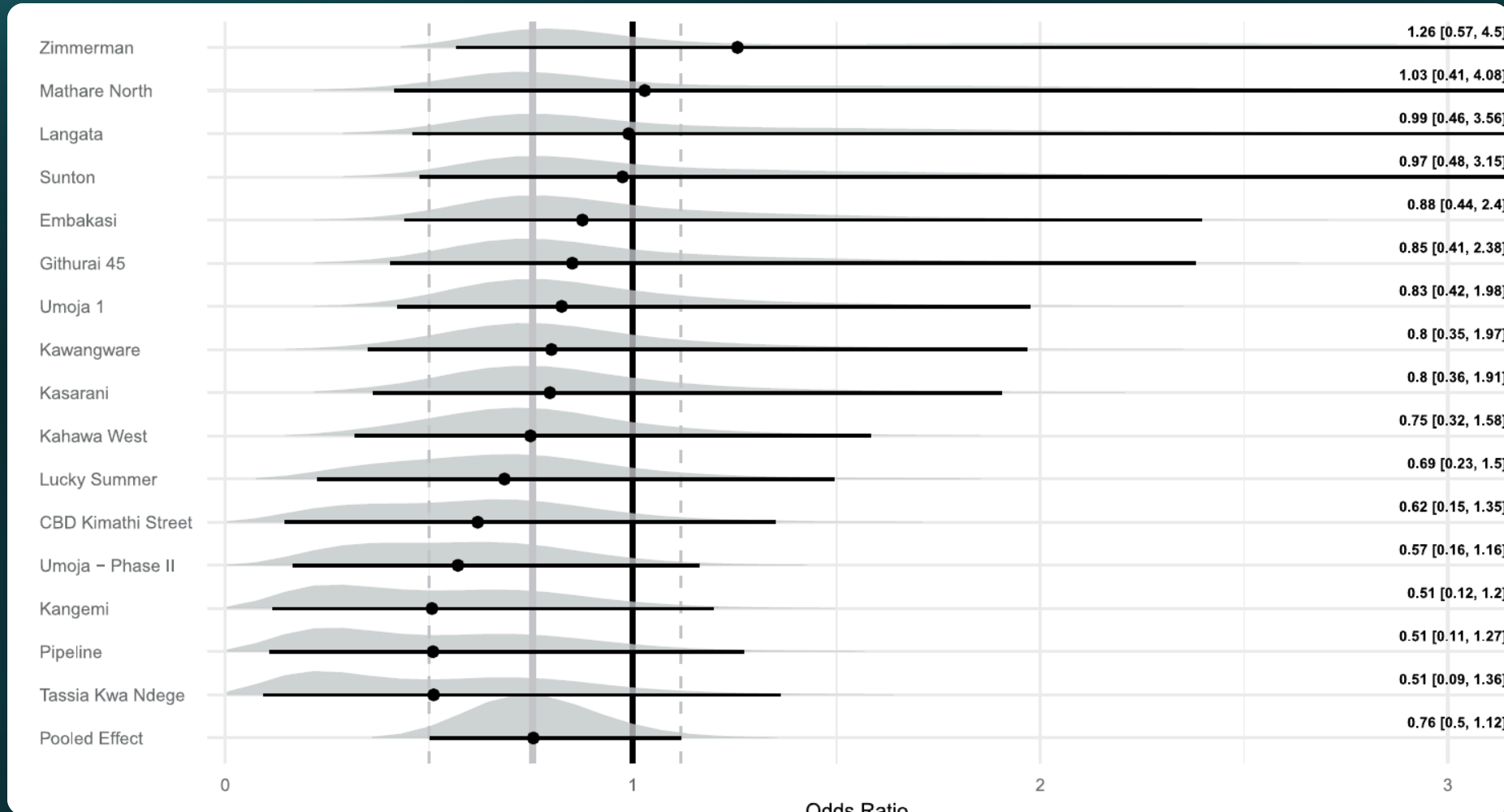
Antibiotics, antimalarials, HTN, malnutrition: no clear effect. Fewer “at risk of T2D” labels in the LLM arm (aOR 0.88, P = 0.023).
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Extended Data Fig. 1 — No subgroup rescues the primary



Age, weekend, sentinel condition, night vs day. Interactions are noisy. Weekend and night samples are tiny.
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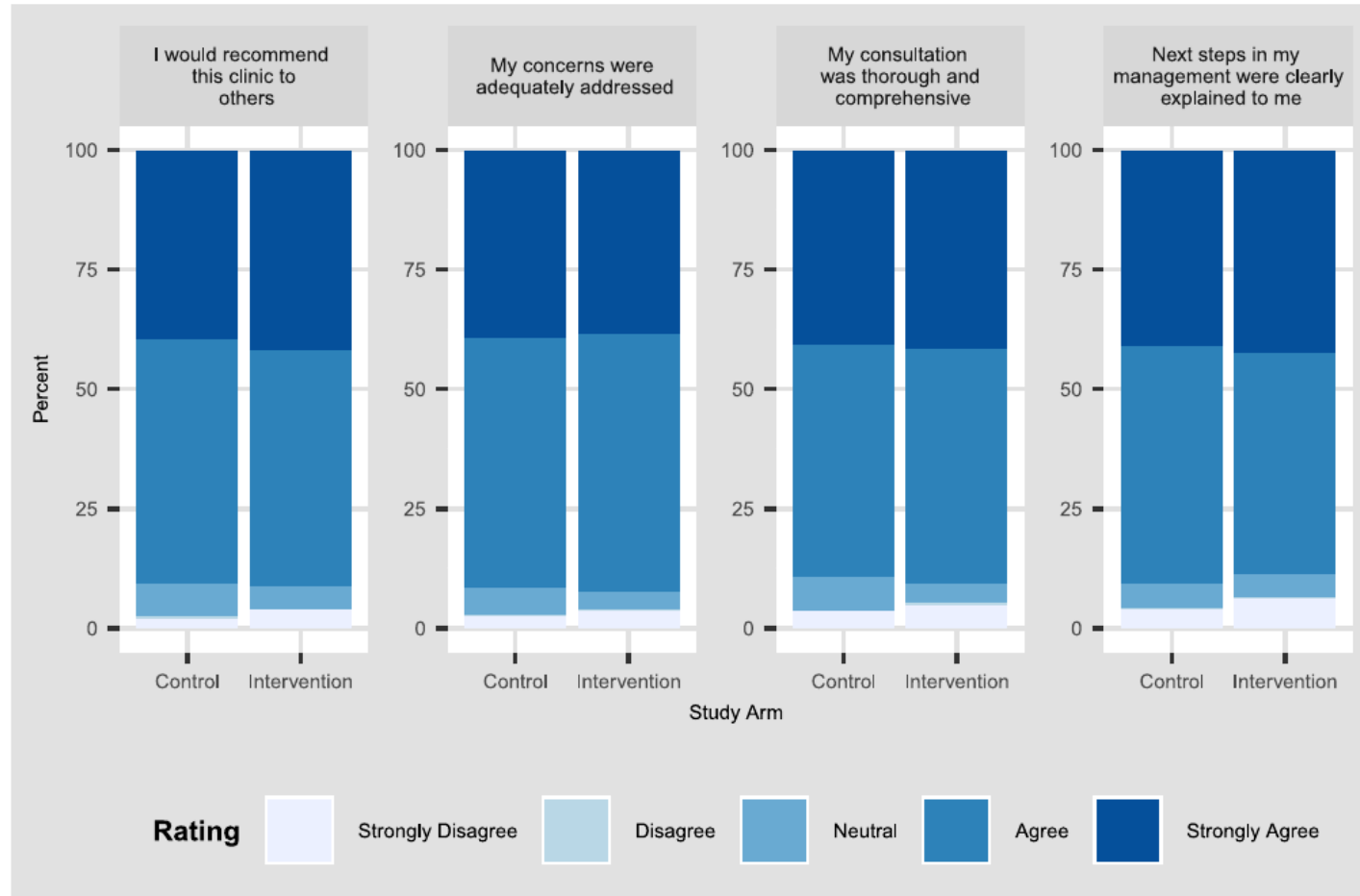
Extended Data Fig. 2 — Same direction, too few events



14 / 16 sites point toward benefit. Pooled OR 0.76 (95% CrI 0.50–1.12). Heterogeneity low ($\tau = 0.22$).

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Extended Data Fig. 3 — Patients did not notice



Median satisfaction 4/5 both arms. Median visit 11 minutes. 33 SAEs (27 hospitalizations, 6 deaths) judged unrelated. Not a powered non-inferiority safety claim.

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INTERPRETATION

Why the primary can be null

1. **Powered for 50% relative reduction.** Observed effect, if any, is a nudge
2. **14-day failure is multi-cause.** Housing, follow-up, the next clinician
3. **Use was optional.** Effectiveness, not enforced efficacy
4. **Penda is already a high-baseline network.** Ceiling effects likely
5. **14 days is short.** Note quality might pay off later — or never

Hospitalization or death would need on the order of **>100,000** patients. What is the right endpoint for a general-purpose copilot in GP?

WHAT I AM TAKING INTO NEXT WEEK

Process moved. Outcomes did not — yet

- Better notes are not nothing. They are not fewer treatment failures
- Override **rate** is the easy metric. Override **quality** is the one in Fig. 2b
- Deskillling is named in the discussion and **not measured**
- One model version, one private urban network, one fortnight of follow-up
- Treat this as a **benchmark**, not a verdict on “AI in medicine”

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YOUR TURN

Paper club questions

1. Next RCT: **which primary endpoint**, and can you actually power it?
2. Documentation up, outcomes flat — success, stepping stone, or the wrong target?
3. How much override is healthy skepticism vs a tool that is too easy to ignore?
4. Public-sector and rural replications **before** anyone scales?

Answers in the comments. That is the club part.

TAKE-HOME

Three lines

Safe*

NO RELATED SAE SIGNAL
AT THIS SCALE

Better notes

DIAGNOSIS, PLAN,
COMPLETENESS

Null primary

14-DAY FAILURE
UNCHANGED

Modest?

LARGE BENEFIT UNLIKELY

*Not a formal non-inferiority safety analysis.

More useful than “AI will transform medicine” or “AI does nothing.”

EPISODE 1 · NOT MEDICAL ADVICE · A CAREFUL READING

See you next week

Paper: Agweyu et al., *Nat Med* (2026) · CC BY 4.0

Figures: unchanged scientific content; ED figures cropped from the PDF

You: record and speak. The agent only preps.

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Next episode: another new, high-impact study that had to survive contact with real patients.

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